

Confounded by Complexity

For decades our output has accelerated in line with our technology. Telephones, computers, robotics, automation, networks, mobiles, laptops, pads, and the internet have played important roles in the advance of science, industry, commerce, and our understanding of the complex world in which we live. However, we have hit a fundamental limit – us! We cannot input, process, and output data any faster, and in many respects we find ourselves confounded by advancing complexity. In short, we need help!

At one level we might see spanners and screwdrivers as muscle amplifiers, and computers as amplifiers of the mind, but computers and their peripherals can be so much more, and for us to access their expanding capabilities we need new interfaces and new ways of working.

Science fiction set public expectation of ‘intelligent machines’ a long time ago with Arthur C Clarke’s 2001 talking computer ‘HAL9000’ followed by ‘Star Trek’ and ‘Terminator’. These movies reinforced the notion that ‘artificial intelligence was in the here and now’ – but in reality it was, and still is, coming!

On the face of it conversing with machines ought to be straightforward engineering problem. After all, accurately recognising and synthesising human speech has been a computing reality for over 25 years, but just try it on your mobile phone in a noisy environment! What goes wrong?

Human language and communication is far more complex and subtle than it first appears, and so is the process of cognition. For example; ‘let us pray’ and ‘let us spray’ sound the same to you and I and a computer, but we are generally aware of the context, and we have understood that they convey entirely different meanings:

Let us pray – we may be in church

Let us spray – we may be in a paint shop, a garden, or a kitchen...

...doing repairs, watering plants, or cleaning a lettuce

Why don’t machines understand? Because we have only recently assembled a

sufficient number of ‘common sense cases’ to afford them some context, and because the computational power required has not been available – but now it is! More importantly, AI is now capable of learning from a vast set of real time inputs in the form of text, audio, video and animations and simulations.

Another subtlety is the lack of sensory devices to give computers continual input and some form of awareness. That too is about to change with the roll out of dedicated sensor networks plus billions of mobile phones equipped with a variety of new capabilities.

So where are we with AI today? The big successes go largely unnoticed and include control systems in industry and aviation, financial trading systems and telecom net-

work control, the design of integrated circuits and complex industrial process and plants. AI systems also populate many engine control systems for our autos, elevators, trains and domestic appliances. On the more esoteric side AI systems now make discoveries in science and invent/innovate in technology and engineering.

And what might we expect soon? For sure voice I/O is going to improve drastically over the next 5 – 10 years, but that HAL9000 style conversation is most likely still 10 – 20 years away. But probably more significantly, the power of search engines will change drastically. Just look at the number of responses I get when I search today:

AI = 1,230,000,000

Artificial Intelligence = 13,900,000

Artificial Intelligence Voice Recognition = 3,600,000

Artificial Intelligence Voice Recognition Cognition = 830,000

All very impressive with response times < 0.2 second, but not a lot of use! I need a system that follows my work, my line of reasoning, and automatically refines searches so that I get the 10 most useful citations. Better still, I need a system that monitors all of my activities and brings relevant documents, messages, calls, history and context to my attention. I also need something anticipatory and relevant to what I am likely to be doing next. But most of all I need a system that can model situations, present me with viable options, and then check out my decisions.

There is one other important feature I am looking forward to – and that is The Truth Engine, The Veracity Checker. So much of the information on the web is wrong or in error. So many times a politician, leader or manager will say something that stretches the bounds of correctness, and an immediate spot check would be all so very powerful.

Science fiction writers mostly predict machines taking over, pushing us aside, and becoming malevolent. My prediction is that we will be happy to let go of a lot of things, like mundane processes and production work, and form a partnership with machines so we can be more creative and productive. 



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